



CONSENT FORM

Computational Principles of Motor Skill: From Action Selection to Adaptation

PRINCIPAL INVESTIGATOR

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TESTING LOCATION

Osborne Centre, Unit 2, Room 203

INVITATION TO PARTICIPATE

You are being invited to take part in this research study because you meet the following inclusion criteria:

- You are a healthy adult between 18 – 35 years old with no known neurological impairments
- You have normal or corrected-to-normal vision
- You are fluent with both written and spoken English
- You are able to comfortably make reaching movements up to 25 cm while holding on to stylus (i.e., pen) or robot handle

In addition, you do not possess or exhibit either of the following:

- Impaired vision that cannot be corrected by glasses or contact lenses
- Hearing or other motor/sensory deficits

OVERVIEW AND PURPOSE

We are interested in better understanding how the human motor system controls reaching movements and utilizes different types of sensory feedback for this purpose. Our studies mimic simple video game-like environments and involve participants making reaching movements to targets. Through control of the visual and somatosensory (i.e., touch and position sense-related) stimuli and careful measurement of movement-related responses, we can advance basic knowledge regarding motor behavior.

PROCEDURES

You are being asked to participate in 1-3 test sessions, with each experimental session requiring approximately 2 hours for completion.

In this experiment, you will be making goal-directed reaches to visually-presented targets. Your primary goal will be to reach as quickly and accurately as possible to the target. In some cases, you may be asked to synchronize your reach with an auditory cue. Depending on the experiment you are assigned to (based on the study being run when you were recruited), you will be tested on either our graphics tablet set-up, in which case you will make reaches while holding on to a stylus (pen), or on our robotic set-up, in which case you will make reaches while holding the handle of the robotic arm.

Depending on the trial type, you may or may not receive visual feedback regarding your hand position. If we utilize the robotic set-up, forces may be applied to your hand during the reach and/or the robot will bring your hand back to the starting position.

A computer will coordinate the equipment to monitor your hand movements. Your movements will be tracked either through the pen placement on the graphics tablet or position encoders embedded within the robotic arm. Computer code will be used to analyze your data. This information will help us understand specific features of your motor performance.

RISKS

The risks involved in participating in this experiment will be minimal. That is, the risks are no greater than the risks in everyday life. You may experience some mental or physical fatigue, due to focusing on the task and making repeated reaches. However, regular breaks will be provided throughout the experiment.

CONFIDENTIALITY

Any personal information resulting from this research study will be kept confidential. All physical documents will be identified only by a numerical code and stored in a locked filing cabinet in the Principal Investigator's research office. All data will be de-identified (i.e., not derived from any personal information such as SIN, DOB, initials, etc.) and coded using a numerical system. Only Dr. Kim and his research assistants will have password-protected access to the encrypted data stored on computers and UBC-approved applications, such as OneDrive. You will not be identified by name in any reports of the completed study. Data will be retained for a minimum of five years post-publication or completion of the study, after which time raw data may be destroyed, including paper documents which will be shredded.

Your confidentiality will be respected. No information that discloses your identity will be released or published. Research records identifying you may be inspected in the presence of the Investigator by representatives of the Natural Sciences and Engineering Research Council of Canada, or the UBC Research Ethics Board for the purpose of monitoring the research. However, no records which identify you by name or initials will be allowed to leave the Investigators' offices.

OPEN ACCES

One of the cornerstone principles of scientific rigor is reproducibility. For this reason, many journals are now requiring researchers to make data associated with a study publicly available. Therefore, only your de-identified quantitative behavioural data (e.g., reaching trajectories, movement velocities) collected during the experiment may be submitted to the publication journal and hosted on a popular data and code hosting website, such as Github (github.com) or Open Science Framework (osf.org). Once the quantitative data are made available, you will not be able to withdraw your data. However, there will be nothing associated with the deposited data that can be linked back to you as all personal identifiers are removed.

REIMBURSEMENT

You will be reimbursed at a rate of \$10.⁰⁰/hr upon completion of your participation.

RIGHTS TO WITHDRAW FROM THE RESEARCH

Your participation is voluntary, and you may withdraw at any time without consequences. You will still be reimbursed for your time. Data collected up to the point of your withdrawal from the study will be kept for data analysis purposes under strict provisions of confidentiality.

COMPENSATION

Signing this consent form in no way limits your legal rights against the sponsor, investigators, or anyone else.

CONTACT FOR INFORMATION ABOUT THE STUDY

All published studies from this lab are publicly available on our lab website (<https://ccmlab.org>). If you have any questions or desire further information with respect to this study, contact Dr. Hyosub Kim at his phone number (604 822 9964), which operates 24 hours a day, seven days a week.

CONTACT FOR COMPLAINTS OR CONCERNS ABOUT THE STUDY

If you have any concerns or complaints about your rights as a research participant and/or your experiences while participating in this study, contact the Research Participant Complaint Line in the UBC Office of Research Ethics at 604-822-8598 or if long distance e-mail RSIL@ors.ubc.ca or call toll free 1-877-822-8598.

FUNDING

This study is funded by the Natural Sciences and Engineering Research Council of Canada (NSERC)..

PARTICIPANT CONSENT

I, _____ (*please print your name*), age _____ (*print your age*) years old, _____ (*print your biological sex*), and _____-handed (*print your dominant hand*) agree to participate in a research project directed by the investigator(s). In so doing, I understand fully all of the following statements:

I have read and understand the procedures used in this study.

I have been given the opportunity to ask questions regarding the procedures and instructions for this study and these questions have been answered fully and to my satisfaction. I understand that I will receive further feedback about the project following my participation.

I understand that the data collected will be used for research publication purposes and kept under strict provisions of confidentiality.

I understand that there are no benefits of joining the research study, as well as the possible risks and discomforts.

I understand that participation in this study is entirely voluntary and that I may refuse to participate, or I may withdraw from the study at any time without any consequences. I understand that signing this consent form in no way limits my legal rights against the sponsor, investigators, or anyone else.

I hereby CONSENT to participate in this study and acknowledge that I will receive a signed and dated copy of the consent form for my own records.

Participant (Print)

Signature

Date